MILITARY SPECIFICATION

INITIATOR, ADAPTION KIT: M50

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of initiator entitled Initiator Adaption Kit: M50 which is used in the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing Aeronautical and Associated Equipment, General Specification for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

ORDNANCE CORPS

8830324	- Initiator, Adaption Kit: M50
EL-8854160	- Equipment Lists

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality assurance Instruction for Reporting Data on Nuclear Weapons

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The initiator shall comply with all requirements specified on the drawing (dwg) 8830324, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SMUPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Continuity and resistance.The continuity and resistance between the following terminals of the initiator shall not exceed the specified resistance when tested as specified in 4.4.1.

From Terminal	To Terminal	Resistance (Ohms)	Test Code Nos.
A	G	0.60 max.	030001
*B	D	0.25 max.	030004
C	f	0.10 max.	
*F	X	0.25 max.	030006

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G	R	0.60 max.	030002
N	R	0.60 max.	030003
*P	U	0.25 max.	030005
S	T	0.10 max.	
*V	W	0.25 max.	030008
*Y	Z	0.25 max.	030009
*c	e	0.25 max.	030007
*g	j	0.25 max.	030010
A	N	1.60 max.	030011

*Relays must be energized for these checks.

3.6.1 Circuit resistance. The circuit resistance between the following terminals of the initiator shall not exceed the specified resistance when tested as specified in 4.4.1.1.

From Terminal	To Terminal	Resistance (Ohms)	Test Code No.
L	M	5600 min.	031001

3.6.2 Accelerometer circuit resistance The circuit resistance between the following normally open accelerometer contacts (when closed) of the initiator shall not exceed the specified resistance when tested as specified in 4.4.1.

From Terminal	To Terminal	Resistance (Ohms)	Test Code Nos. 1	2	3
J	H	0.3 max.	032001	032004	032007
J	K	0.3 max.	032002	032005	032008
J	E	0.3 max.	032003	032006	032009

3.7 Diode characteristics. Forward voltage drop across the diodes and the reverse voltage drop across a 100 ± 1 kilohm (k) series resistor shall be as specified when tested as specified in 4.4.2.

Pos. Lead To Term.	Neg. Lead To Term.	Applied Forward Current	Max. Fwd. Volt. Drop	Applied Reverse Voltage	Max. Volt. Drop Across Series Res.	Test Code Nos.
B	J	500 ± 25 ma	1.2			040001
X	J	500 ± 25 ma	1.2			040002
U	S	400 ± 20 ma	1.0			040003
W	S	400 ± 20 ma	1.0			040004
J	B & X			200 ± 10	0.3 (5)	040005
S	U & W			200 ± 10	0.3 (5)	040006

(Figure in parenthesis in last column is max. voltage drop at + 160°F.)

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3.8 The capacitor characteristics between the pins listed below shall be capable of retaining a charge potential of 40 to 52 volts for 60 ± 5 seconds after the voltage source is removed when tested as specified in 4.4.3. Except at low temperature, the capacitor charge retention requirement shall be 33 to 55 volts.

Positive Lead to Terminal	Negative Lead to Terminal	Retained Charge Potential	Test Code Nos.
B	C	40 to 52 Volts	050001
X	C	40 to 52 Volts	050002
U	C	40 to 52 Volts	050003
W	C	40 to 52 Volts	050004
a	C	40 to 52 Volts	050005
b	C	40 to 52 Volts	050006

3.9 Insulation resistance. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all other terminals listed below and case when tested as specified in 4.4.4.

A,C,D,E,F,H,K,L,P,V,Z,e,g,j,Y and c

Unused terminals d and h shall be connected together and tested as one terminal.

3.10 Integrating accelerometer operation. The accelerometer shall operate as specified for functional operation 3.10.1 and for functional threshold 3.10.2 when tested as specified in 4.4.5.

3.10.1 Functional operation. The operational g-second product of each accelerometer shall be as specified below when tested as specified in 4.4.5.1.

From Terminal	To Terminal	G-Sec Requirement	Test Code Nos. 1	2	3
J	H	8.0 ± 2.75	071001	071004	071007
J	K	8.0 ± 2.75	071002	071005	071008
J	E	8.0 ± 2.75	071003	071006	071009

3.10.2 Functional threshold. The functional threshold of each accelerometer shall be 2.9 ± 0.6 gravity units (g's) when tested as specified in 4.4.5.2.

3.11 Environmental testing. The initiator shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The initiator shall also comply with 3.6 through 3.10 before and after each environmental test. Testing shall be as specified in 4.4.6.

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3.11.1 High temperature. The initiator shall withstand exposure to a temperature of 160 ± 5 degrees Fahrenheit (°F.) without damage and while at this temperature shall meet the requirements of 3.7, 3.8, 3.9 and 3.10 when tested as specified in 4.4.6.1. The initiator shall meet the requirements of 3.6 and 3.6.1 after completion of high temperature test at 80 ± 20°F.

3.11.2 Low temperature. The initiator shall withstand exposure to a temperature of -65° ± 5°F. without damage and while at this temperature, shall meet the requirements of 3.7, 3.8, 3.9 and 3.10 when tested as specified in 4.4.6.2. The initiator shall meet the requirements of 3.6 and 3.6.1 after completion of low temperature test at 80 ± 20°F.

3.11.3 Humidity. The initiator shall withstand exposure to humidity without evidence of damage affecting functioning when tested as specified in 4.4.6.3. The assembly shall meet the requirements of 3.6, 3.6.1, 3.7, 3.8, 3.9 and 3.10 after the conclusion of the test.

3.11.4 Transportation vibration. The initiator shall withstand vibration for 15 minutes in each of three mutually perpendicular axes, as shown in Figure 1, at the frequencies and amplitudes specified in Table I without evidence of damage or affecting function when tested as specified in 4.4.6.4. The frequency shall be varied uniformly from 8 to 500 to 8 Hertz (Hz) in 15 minutes. After being subjected to the specified vibration, the initator shall meet the requirements of 3.6, 3.6.1, 3.7, 3.8, 3.9 and 3.10.

Table I

Hertz (Hz)	Cycle Time	Constant Acceleration of Double Amplitude
8-55-8	7.5 minutes	0.060 ± 0.006 inch
55-500-55	7.5 minutes	10 ± 1 g

3.11.5 Operational vibration. The initiator shall withstand without evidence of damage or affecting functioning a programmed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration without evidence of damage or affecting function when tested as specified in 4.4.6.5. The total time of test shall be 210 seconds; 70 seconds in each of the three perpendicular axes shown in Figure 1. The complex vibration test will consist of the following five sections, programmed continuously:

1. White Gaussian noise at a level of 22 gravity units (g) root mean square (RMS) band-limited between 15 and 1500 Hz for one (1) second.

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2. A 4 g rms swept sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at a rate increasing directly with frequency, one such sweep taking the full 15 seconds.

3. An 8 g rms swept sine wave superimposed on 15 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.

4. A 4 g rms sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 38 seconds. The sine wave shall be swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.

5. White Gaussian noise at a level of 22 g rms band-limited between 14 and 1500 Hz for one (1) second. Following the vibration test, the initiator shall meet the requirements of 3.6, 3.6.1, 3.7, 3.8, 3.9 and 3.10.

3.11.6 Shock. The initiator shall withstand shocks of 30 + 5 g's three times in each direction along the three mutually perpendicular axes shown in Figure 1 for a total of 18 shocks when tested as specified in 4.4.6.6. Following the tests, the assembly shall meet the requirements of 3.6, 3.6.1, 3.7, 3.8, 3.9 and 3.10.

3.12 Soldering. All soldering shall comply with MIL-STD-1460.

3.13 Test Codes (see 3.5). In addition to the normal inspection records, the contractor shall provide data feed back on punch cards in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and code numbers, are specified in the applicable paragraphs in this section.

3.14 Workmanship.

3.14.1 Plastic and metal characteristics. All components shall be free from crakcs, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.14.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.14.3 Burrs. Parts shall be free from burrs.

3.14.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.14.5 Cleaning. The cleaning method shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent.

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3.14.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8830324) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II

First article inspections and tests

Inspection or Test	Dest	Non-Dest	Testing Sequence 1	2	3
1. Classification of characteristics (see 4.3.2.1)		X	X	X	X
2. Continuity and resistance (see 3.6)		X	X	X	X
3. Circuit resistance (see 3.6.1)		X	2	2	2

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4.	Insulation resistance (see 3.9)		X	3	3	3
5.	Diode characteristics (see 3.7)		X	4	4	4
6.	Capacitor characteristics (see 3.8)		X	5	5	5
7.	Integrating accelerometer operation (see 3.10)		X	6	6	6
8.	Transportation vibration (see 3.11.4)	x		7	7	
9.	High temperature (see 3.11.1)	X		9	8	9
10.	Low temperature (see 3.11.2)		X		9	7
11.	Operational vibration (see 3.11.5)	X				8
12.	Shock (see 3.11.6)	X		8	11	
13.	Humidity (see 3.11.3)	X		10	10	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Initiator (prior to potting) (dwg. 883024)

Categories	Characteristics	Method of Inspection
Critical:		
	1. Incorrect soldering (see 3.12)	Visual
	2. Loose connection	Visual/Manual
	3. Damaged or missing component	Visual
	4. Damaged wire insulation	Visual
	5. Parallelism of brackets out of tolerance	Gage
Major:		
	101. Objective evidence relative to material missing or improper	Visual
Minor:		
	201. Mis-orientation of connector	Visual
	202. Case cracked or damaged	Visual
	203. Evidence or poor workmanship (see 3.14)	Visual
	204. No chamfer on left bracket	Gage/Visual

4.3.2.1.2 Initiator (dwg. 8830324)

Categories	Characteristics	Method of inspection
Critical:		
	1. Damaged, loose, or missing sockets on connector	Visual
	2. Inability of connector assembly to fit mating part	Gage

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Major:

101.	Improper location of mounting holes	Gage
102.	Maximum external dimensions of assembly below brackets out of tolerance	Gage
*103.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/32 inch in depth relative to the adjacent sockets)	Visual/Gage
104.	Diameter of mounting holes out of tolerance	Gage
105.	Objective evidence relative to material missing or improper (see 3.1)	Visual
106.	Evidence of poor workmanship (see 3.14)	Visual

Minor:

201.	Insufficient or excessive potting	Visual
202.	Incorrect or illegible marking	Visual
203.	Bare spot in protective coating, except for slight scratches	Visual
204.	Serialization improper or missing (see 3.4)	Visual

*In the event socket inserts do not appear to be within 1/32 inch in depth relative to the adjacent sockets, gage (see 4.3.4) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.032 inch maximum from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

4.3.2.1.3 Packing and marking for storage and shipment (see dwg. 8830324 and section 5)

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
101.	Preservation, packing material and packaging not in accordance with applicable drawing	Visual
102.	Marking not in accordance with applicable drawing	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.14)	Visual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity and resistance	Major
b. Circuit resistance	Major
c. Diode characteristics	Major
d. Capacitor characteristics	Major
e. Insulation resistance	Major
f. Integrating accelerometer operation	Major
g. Low temperature	Major
h. Transportation vibration	Major
i. Shock	Major

Of these tests, (h) and (i) are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Continuity and resistance, circuit resistance, diode characteristics, capacitor characteristics, insulation resistance and integrating accelerometer operation. Each initiator in the lot shall be tested in accordance with 4.4.1 through 4.4.5. If any item fails to comply with 3.6 through 3.10, the item shall be rejected and removed from the lot.

4.3.3.2 Low temperature, transportation vibration and shock. A sample shall be selected at random in accordance with Table V. The sample shall be tested in accordance with 4.4.6.2, 4.4.6.4 and 4.4.6.6. If any item fails to comply with 3.11.2, 3.11.4 and 3.11.6, the lot shall be rejected.

Table V

Sample sizes for testing

Number Required to be Delivered	Sample Size for No Failures
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7
1301-3200	10

Lot Size Required = Number Required to be Delivered + Sample Size.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854160). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity and resistance. The continuity and resistance shall be measured between the terminals listed in 3.6 with a four wire resistance measuring instrument. Where necessary energize relays by applying 115 ± 10 volts direct current (vdc) between terminals L and M. The resistance of circuits containing normally open accelerometer contacts shall be measured only while the initiator is in the g-field. Record values (see 3.13).

4.4.1.1 Circuit resistance. The circuit resistance between the terminals listed in 3.6.1 shall be measured with a digital ohmmeter or a wheatstone bridge. Record values (see 3.13).

4.4.2 Diode characteristics. Diodes shall be tested for forward voltage drop when current specified in 3.7 is applied in the forward direction. Reverse current shall be determined by the voltage drop across a 100 k ± 1k ohm series resistor when voltage is applied in the reverse direction. The minimum input impedance of the voltmeter to monitor the voltage drop across this resistor shall be 9 megohms. The duration of the tests in either the forward or reverse direction shall not be more than 15 seconds. Record values (see 3.13).

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4.4.3 Capacitor characteristics. The capacitance between terminals listed in 3.8 shall be tested for capability to hold charge. Connect a power supply across each pair of terminals and charge capacitors to 55 + 1 - 0 vdc for 130 +0 -10 seconds. Disconnect assembly from supply and after 60 + 5 seconds connect to a voltmeter with 10 megohm impedance min. Record voltage reading (see 3.13). Repeat the above for each pair of terminals.

4.4.4 Insulation resistance. The insulation resistance of the initiator shall be measured between the terminals specified in 3.9 using an insulation tester having an output of 500 + 50 vdc and a current limited to 5 milliamperes maximum.

4.4.5 Integrating accelerometer operation. For tests 4.4.5.1 and 4.4.5.2, the following terminals of the initiator shall be monitored.

Accelerometer	Normally Closed Terminals	Normally Open Terminals
1	G and R	J and H
2	N and R	J and K
3	A and G	J and E

4.4.5.1 Functional operation. The initiator shall be mounted securely on a centrifuge in a suitable jig. The unit shall be positioned by the jig such that acceleration of the centrifuge will not cause the normally closed (N.C.) monitor circuits of the accelerometers to open. When the acceleration on the component is at 3.5 ± 0.1 g's, the orientation of the unit shall be changed such that the centrifugal acceleration will be along the longitudinal axis of the component in the direction indicated by the arrow in Figure 1 within ± 5° of arc. The time required to perform the final 90° of this change or orientation shall not exceed 0.250 seconds. When determining g-second product, the actual acceleration shall be multiplied by the measured time between N.C. circuit opening and N.O. circuit closing. The N.C. circuit shall be closed prior to initiating each test sequence. Perform the test three times and record the g-second measurements (see 3.13). The radius arm of the centrifuge shall be the distance from the centrifuge center to a point 1.375 ± 0.001 inches from the surface opposite the connector. Upon closure of the normally open accelerometer contacts and while maintaining the same centrifuge rpm, measure the resistance (as indicated in 4.4.1) of the following circuits.

Accelerometer	From Terminal	To Terminal	Resistance (OHMS)
1	J	H	0.3 Max.
2	J	K	0.3 Max.
3	J	E	0.3 Max.

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4.4.5.2 Functional threshold. The unit shall be mounted securely in the jig used for functional operation oriented such as shown in Figure 1 within ± 5° of arc. The centrifuge shall be brought slowly from rest to that speed which will subject the unit to 2.5 g's. The N.C. circuit shall remain closed. The centrifuge speed shall then be slowly increased to that speed which will subject the unit to 3.5 g's. The N.C. circuit shall have opened and the N.O. circuit shall have closed. Upon closure of the normally open accelerometer contacts and while maintaining the same centrifuge rpm, measure the resistance (as indicated in 4.4.1) of the following circuits.

Accelerometer	From Terminal	To Terminal	Resistance (OHMS)
1	J	H	0.3 Max.
2	J	K	0.3 Max.
3	J	E	0.3 Max.

4.4.6 Environmental testing (see 3.11)

4.4.6.1 High temperature. The initiator shall be placed in a test chamber. The chamber's temperature shall be raised and stabilized at 160 ± 5°F. (see 6.4). After the control unit has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.7, 3.8, 3.9 and 3.10. After completion of the high temperature tests when the temperature of the initiator has returned to ambient (80 ± 20°F), the initiator shall be tested for compliance with the requirements of 3.6 and 3.6.1.

4.4.6.2 Low temperature. The initiator shall be placed in a test chamber. The chamber's temperature shall be lowered and stabilized at -65 + 5°F (see 6.4). After the initiator has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.7, 3.8, 3.9 and 3.10. After completion of the low temperature tests when the temperature of the initiator has returned to ambient (80 ± 20°F), the initiator shall be tested for compliance with the requirements of 3.6 and 3.6.1.

4.4.6.3 Humidity. The initiator, with its connector covered with a potted mating connector, shall be placed in a humidity test chamber and shall be subjected to the humidity test as specified in MIL-E-5272, 4.4, procedure 1, except that the number of cycles shall be three. All temperatures shall be maintained at ± 5°F. The pH value shall be adjusted by the use of chemically pure sodium hydroxide. Upon completion of the last cycle, the initiator shall be removed from the test chamber and allowed to return to ambient

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temperature (80 ± 20°F.) without forced cooling. After returning to ambient temperature, the initiator may be wiped and then blown with filtered, compressed air for a period of five minutes, and then shall be tested for compliance of 3.6, 3.6.1, 3.7, 3.8, 3.9 and 3.10.

4.4.6.4 Transportation vibration. The initiator shall be mounted on a vibration machine and vibrated for 15 minutes in each of three mutually perpendicular axes, as shown in Figure 1, at the frequencies and amplitudes specified in 3.11.4. Total test time for each initiator shall be 45 minutes. Following the vibrations, the initiator shall be tested for the requirements of 3.6, 3.6.1, 3.7, 3.8, 3.9 and 3.10.

4.4.6.5 Operational vibration. The initiator shall be mounted on a vibration machine and vibrated in three mutually perpendicular axes, as shown in Figure 1, using a standard noise or complex wave tape. As a minimum the vibration levels shall be observed on the shake table as near the supporting bracket as is practical. The tape shall have the characteristics listed in the following table.

Time (sec)	Type of Signal	Ratio of Test Level To Calibration Level	RMS g Level
0-20	Noise (calibration)(1)	1.	Set gain to give 2 g rms
20-30	None	-	-
30-31	Noise (1)	11.	22.
31-46	Noise (1) plus sinusoid (2)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
46-61	Noise (1) plus sinusoid (3)	8.5	17.
	Noise alone	7.5	15.
	Sinusoid alone	4.	8.
61-99	Noise (1) plus sinusoid (4)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
99-100	Noise (1)	11.	22.

(1) Band limited (15-1500 Hz) white Gaussian noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

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For creating the random noise, the output of a General Radio random noise generator, or its equivalent, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring the correct noise bandwidth, a filter with an asymptotic slope of at least 24 Db per octave and 3 Db points at 15 and 1500 Hz shall be considered acceptable. Following the vibration, the initiator shall be tested for the requirements of 3.6, 3.6.1, 3.7, 3.8, 3.9 and 3.10.

4.4.6.5.1 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire vibration system shall be as flat as is consistent with the present state of the art. Plus or minus 3 Db or better over the band from 15 to 1500 Hz shall be considered as an objective in this respect. After proper equalization of the system the standard noise tape is applied. The over-all gain of the shake system is then adjusted so that with the noise calibration signal, the g level listed in 4.4.6.5 is obtained. With the same gain setting, the tape is run for the remainder of the test and g levels obtained will conform to the levels listed in the table of 4.4.6.5.

4.4.6.6 Shock. The initiator shall be mounted in a shock device and subjected to a total of 18 shocks of 30 ± 5 g, three in each direction along each of three mutually perpendicular axes, as shown in Figure 1. The time duration of each shock shall be 11 ± 2 milliseconds (ms) as measured between points which are 10 percent of the peak values. The time to peak shall be 5.5 ± 1 ms. Upon the completion of the shocks the initiator shall be tested for the requirements of 3.6, 3.6.1, 3.7, 3.8, 3.9 and 3.10.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The initiator shall be packaged in accordance with dwg 8830324 using the specified moisture barrier.

5.1.2 Level C. The initiator shall be packaged in accordance with dwg 8830324 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The initiator shall be packed in accordance with dwg 8830324.

5.3 Marking. Marking shall be as specified in accordance with dwg 8830324.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.6, 3.6.1, 3.6.2, 3.7, 3.8, 3.10 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2071 Revision 1 dated 4 June 1965.

Custodian:
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Preparing activity:
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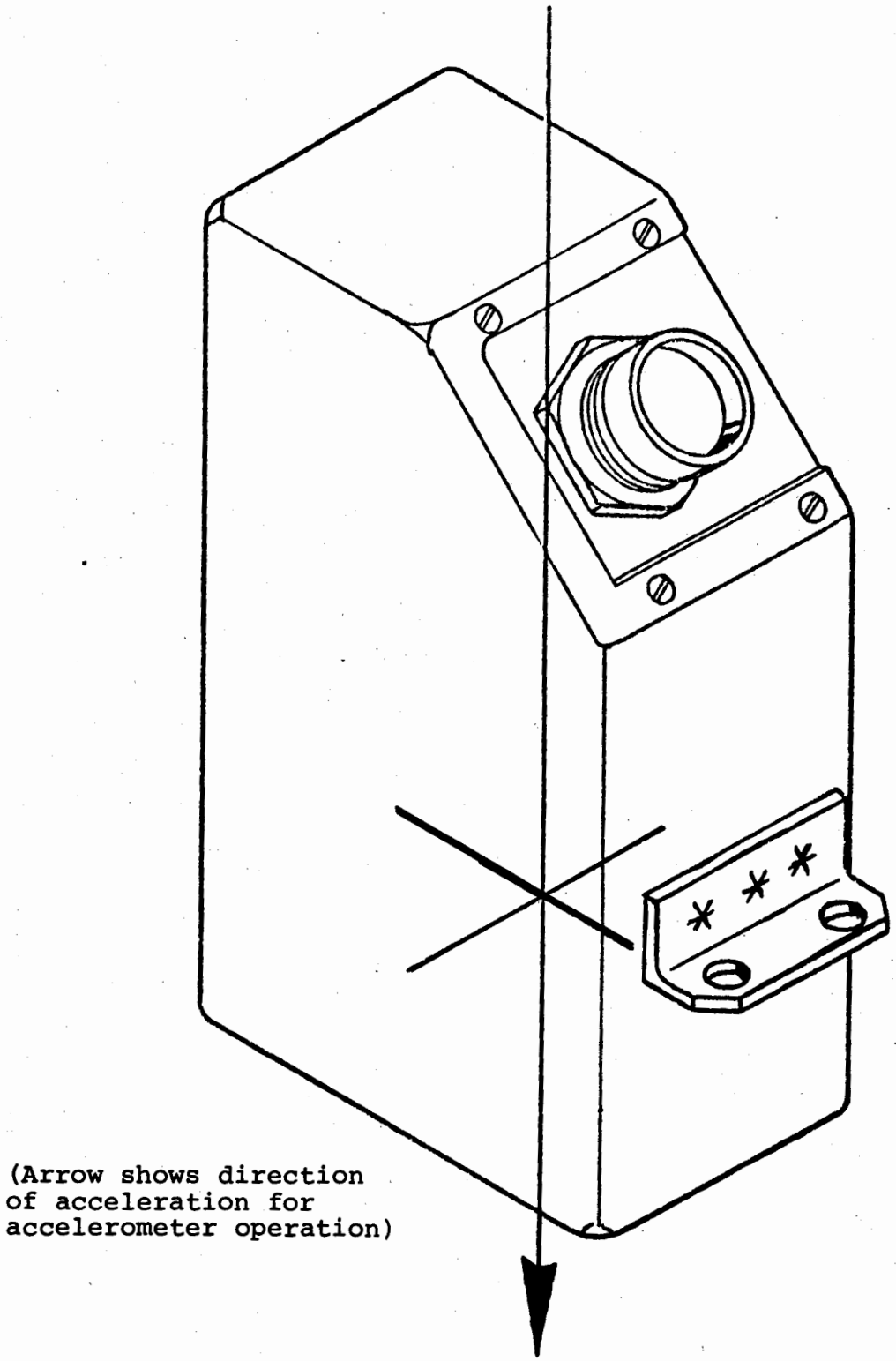


Figure 1 - Axes for vibration, shock and acceleration of initiator

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